

RAAM
Regime-Aware Asset Management

QWIM Market Regime Detection and Regime-Aware Portfolio Allocation

Medium–Long Term Wealth Management Application

Final Presentation

FE800

Stevens Institute of Technology

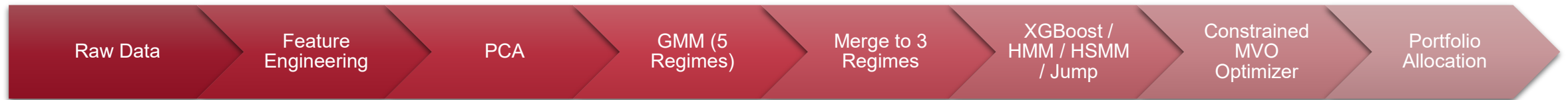
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Agenda

- RAAM Framework Overview
- Regime Identification
- Overall Results
- Annual Performance
- Annual Alpha
- Regime Alpha Attribution
- Turnover & Transaction Costs
- Drawdown Analysis
- Risk Metrics
- Risk Profile Comparison
- 2022 Limitation
- Dashboard Demo
- Conclusion & Recommendation



RAAM Framework Overview



Data

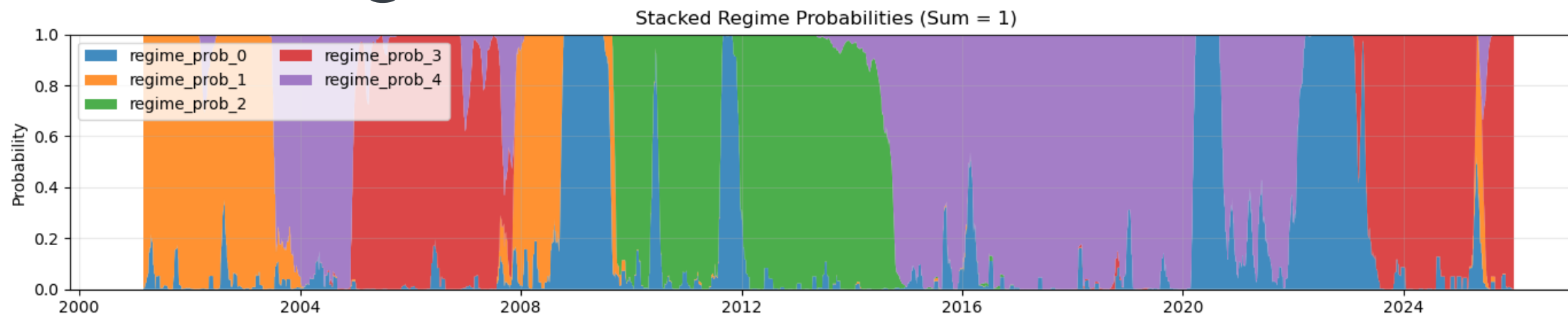
Dec 1999 – Dec 2025 | Bloomberg + Fama-French | 49 Features across 12 buckets

Benchmarks

Static 55/40/5 | MV No-Regime | Walk-Forward 2019–2025



Economic Characterization of Identified Market Regimes



Regime	Name	SPX Return	VIX	Credit Spread	Yield Curve	Eq-Bond Corr
0	Crisis / Panic	↓ Negative	Very High (~31)	Wide	↘ Flat/Low	Negative
1	Recession / Credit Stress	↓ Negative	High (~25)	Highest	↗ Steepening	Very Negative
2	Expansion / Risk-On	↑ Strong Positive	Moderate (~17)	Tight	Steep	Negative
3	Late Cycle / Policy Tightening	↑ Positive	Low (~14)	Tightest	Flat	Positive
4	Goldilocks / Stable Growth	↑ Positive	Low (~16)	Tight	Normal	Negative

Overall Results

All Models vs Benchmarks (2019–2025, Moderate Profile, \$100k Initial)

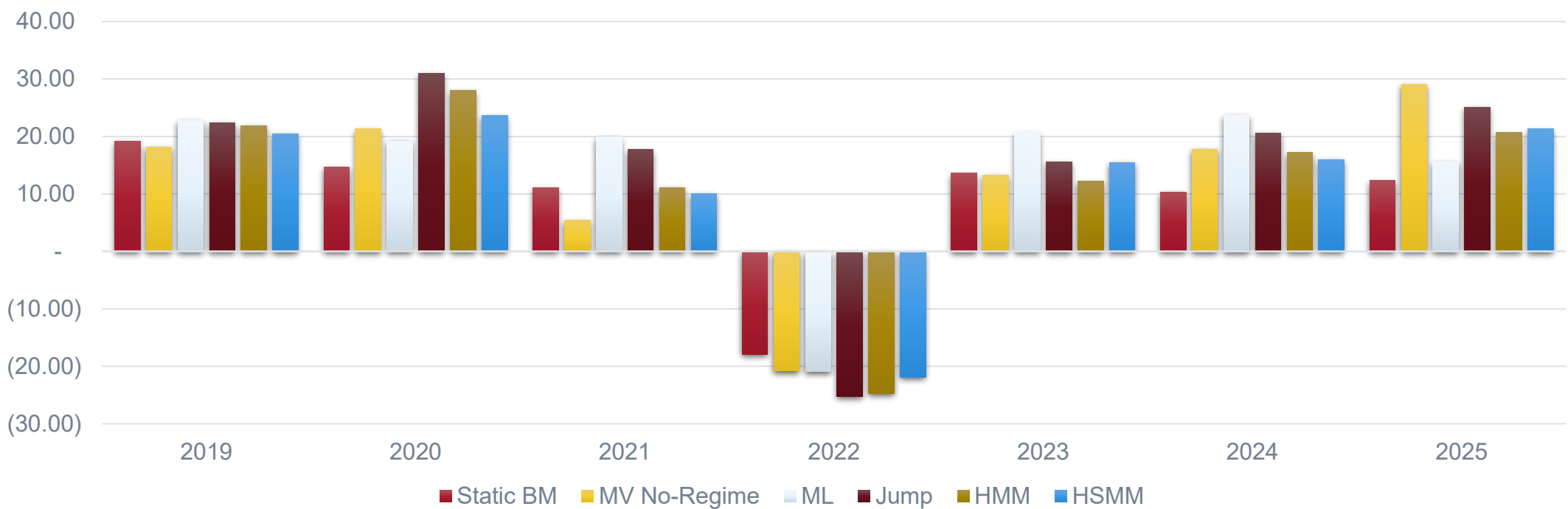
Metric	Static BM	MV No-Regime	ML	Jump	HMM	HSMM
Initial (\$)	\$ 100,000	\$ 100,000	\$ 100,000	\$ 100,000	\$ 100,000	\$ 100,000
Final (\$)	\$ 175,379	\$ 206,240	\$ 239,406	\$ 246,396	\$ 207,187	\$ 208,289
Total Return	75%	106%	139%	146%	107%	108%
Ann. Return	8.4%	10.9%	13.3%	13.8%	11.0%	11.1%
Ann. Vol	12.4%	11.9%	12.4%	12.8%	13.2%	14.7%
Sharpe	0.71	0.93	1.07	1.07	0.86	0.79
Sortino	0.66	0.90	1.00	1.02	0.80	0.74
Calmar	0.37	0.41	0.54	0.46	0.39	0.39
Max Drawdown	-22.40%	-26.70%	-24.80%	-29.70%	-28.50%	-28.10%
Win Rate (Mo)	66.7%	65.5%	66.7%	70.2%	64.3%	65.5%
Switches	-	-	34	8	31	25



Annual Performance

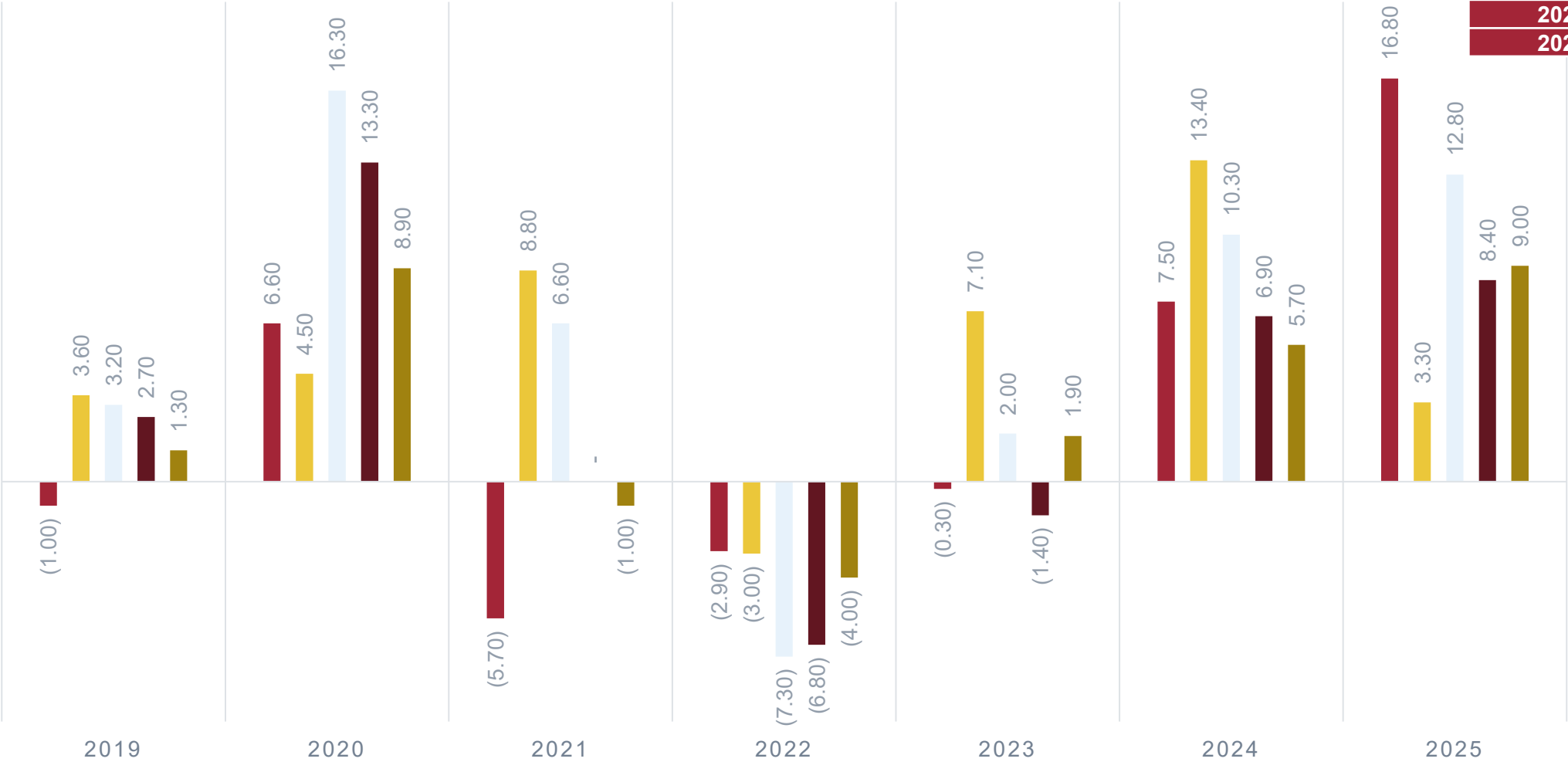
Annual Performance Breakdown
— All Models vs Benchmarks

Year	Static BM	MV No-Regime	ML	Jump	HMM	HSMM
2019	19.20	18.20	22.80	22.40	21.90	20.50
2020	14.70	21.30	19.20	31.00	28.00	23.60
2021	11.10	5.40	19.90	17.70	11.10	10.10
2022	(17.90)	(20.80)	(20.90)	(25.20)	(24.70)	(21.90)
2023	13.60	13.30	20.70	15.60	12.20	15.50
2024	10.30	17.80	23.70	20.60	17.20	16.00
2025	12.30	29.10	15.60	25.10	20.70	21.30



Annual Alpha

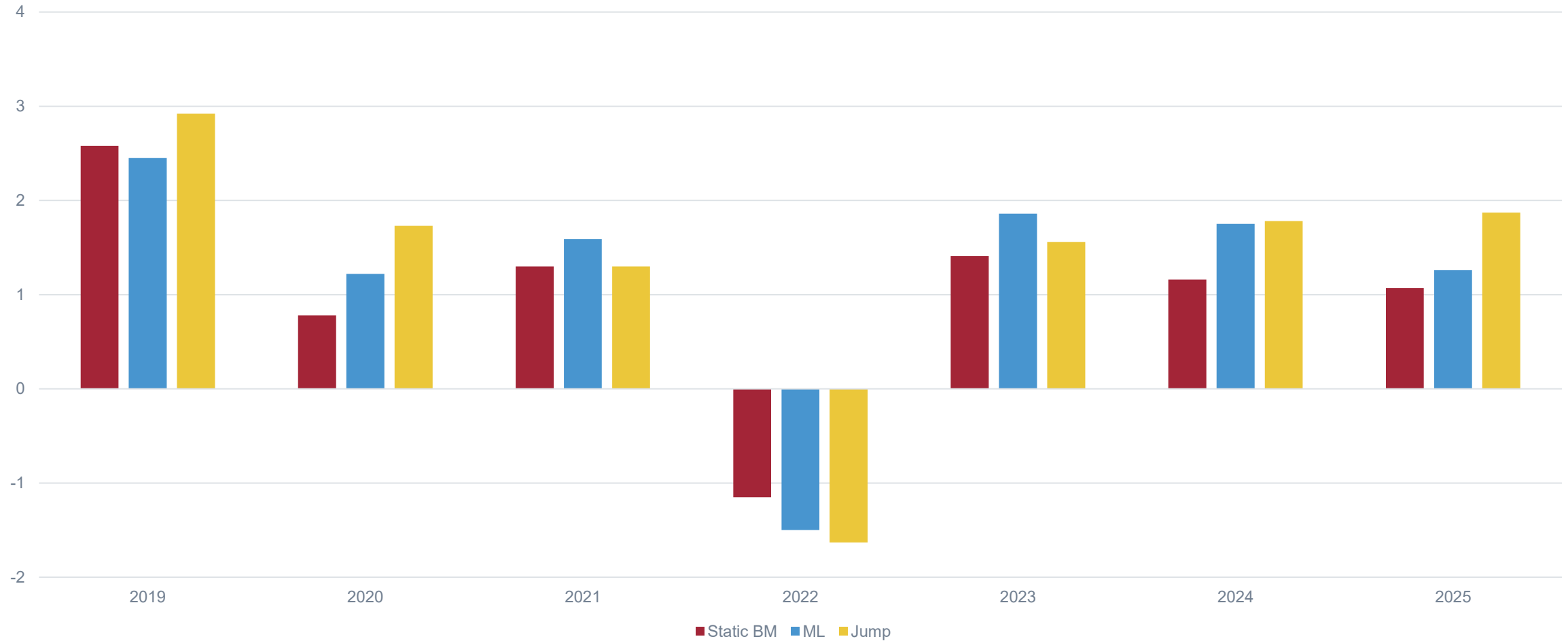
MV No-Regime ML Jump HMM HSMM



Year	Highest Alpha
2019	ML
2020	Jump
2021	ML
2022	Static BM
2023	ML
2024	ML
2025	MV

Annual – Risk Adjusted Return

Annual Sharp Ratio



Regime Alpha Attribution

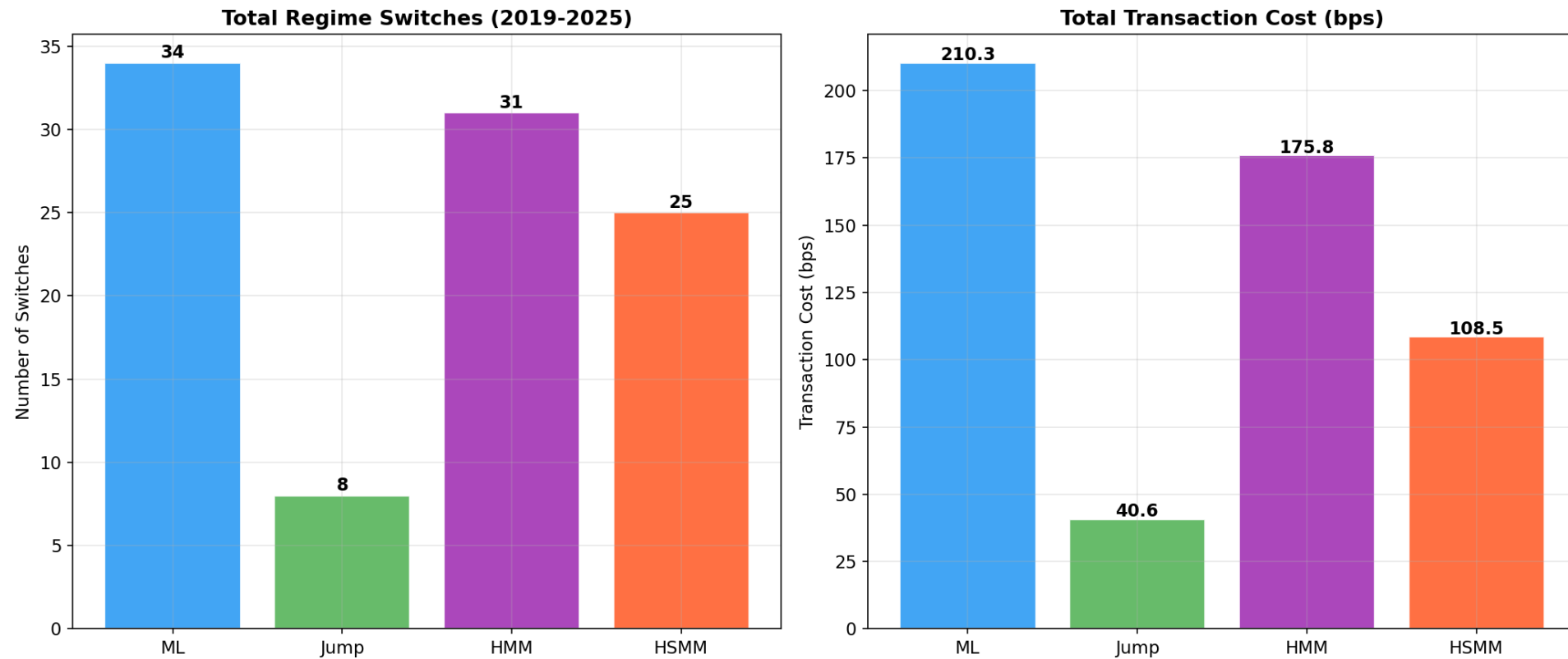
ML	Days	% Time	Regime Ret	Bmark Ret	Alpha	Regime Sharpe	Bmark Sharpe	Change Sharpe	Sortino	MaxDD
Bull	988	56.10%	15.72%	8.20%	7.52%	1.328	1	0.328	1.242	-10.10%
Transition	467	26.50%	8.50%	4.48%	4.02%	0.704	0.42	0.284	0.663	-17.50%
Bear	305	17.30%	13.15%	15.24%	-2.09%	0.91	0.777	0.132	0.844	-12.90%

CJ	Days	% Time	Regime Ret	Bmark Ret	Alpha	Regime Sharpe	Bmark Sharpe	Change Sharpe	Sortino	MaxDD
Bull	310	17.60%	22.86%	13.83%	9.03%	1.385	1.429	-0.044	1.33	-12.10%
Transition	1342	76.20%	10.93%	5.03%	5.90%	0.95	0.499	0.451	0.915	-25.40%
Bear	108	6.10%	25.20%	39.03%	-13.83%	1.409	1.342	0.067	1.18	-10.90%



Turnover & Transaction Costs

A transaction cost of 5 basis points per trade was applied each time the portfolio changed allocation



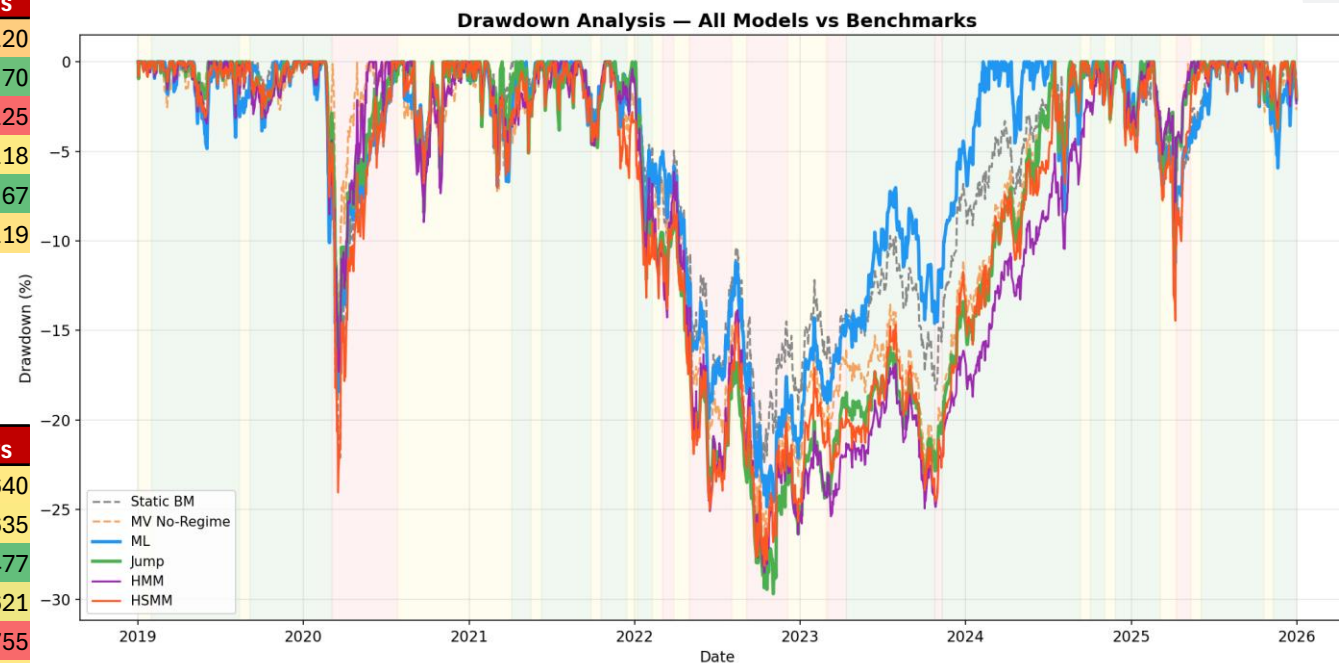
Drawdown Analysis

COVID-19 Drawdown

Model	Peak	Trough	Recovery	Depth	DD Days	Rec Days
Static BM	2/20/2020	3/23/2020	7/21/2020	-22.10%	32	120
MV	2/21/2020	3/18/2020	5/27/2020	-16.90%	26	70
ML	2/19/2020	3/19/2020	7/22/2020	-18.40%	29	125
Jump	2/19/2020	3/19/2020	7/15/2020	-16.90%	29	118
HMM	2/19/2020	3/20/2020	5/26/2020	-17.30%	30	67
HSMM	2/19/2020	3/18/2020	7/15/2020	-24.00%	28	119

2022 Rate Shock Drawdown

Model	Peak	Trough	Recovery	Depth	DD Days	Rec Days
Static BM	11/8/2021	10/14/2022	7/15/2024	-22.40%	340	640
MV	11/9/2021	10/20/2022	7/16/2024	-26.70%	345	635
ML	11/19/2021	10/20/2022	2/9/2024	-24.80%	335	477
Jump	1/3/2022	11/3/2022	7/16/2024	-29.70%	304	621
HMM	11/8/2021	10/14/2022	11/7/2024	-28.50%	340	755
HSMM	11/8/2021	10/14/2022	7/16/2024	-28.10%	340	641



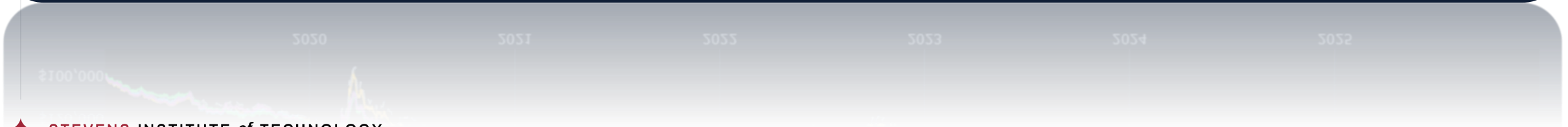
Risk Metrics

Model	Ann. Vol	VaR (95%)	CVaR (95%)	Skewness	Kurtosis
Static BM	12.43%	-1.12%	-1.85%	-0.418	12
MV No-Regime	11.88%	-1.14%	-1.74%	-0.063	6.362
ML	12.44%	-1.25%	-1.87%	-0.309	3.033
Jump	12.82%	-1.26%	-1.91%	-0.112	5.097
HMM	13.15%	-1.34%	-2.04%	-0.3	3.922
HSMM	14.72%	-1.39%	-2.20%	-0.266	7.343



Risk Profile – Moderate

Asset	Weight
Equity	55%
Bond	40%
Gold	5%



Risk Profile – Conservative

Asset	Weight
Equity	30%
Bond	60%
Gold	10%



Risk Profile – Aggressive

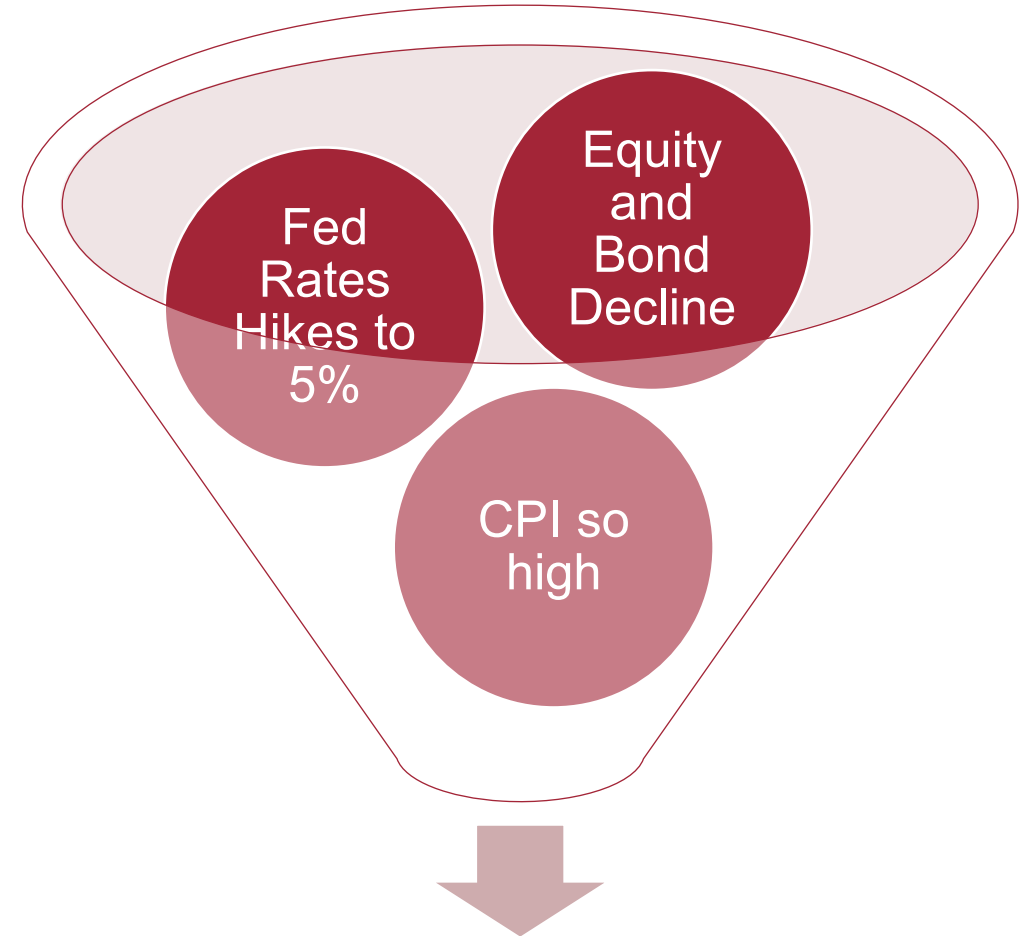
Asset	Weight
Equity	85%
Bond	10%
Gold	5%



Key Limitation

The 2022 Structural Break

Model	Return	Alpha vs BM
Static BM	-17.9%	—
ML	-20.9%	-3.0%
Jump	-25.2%	-7.3%
HMM	-24.7%	-6.8%
HSMM	-21.9%	-4.0%



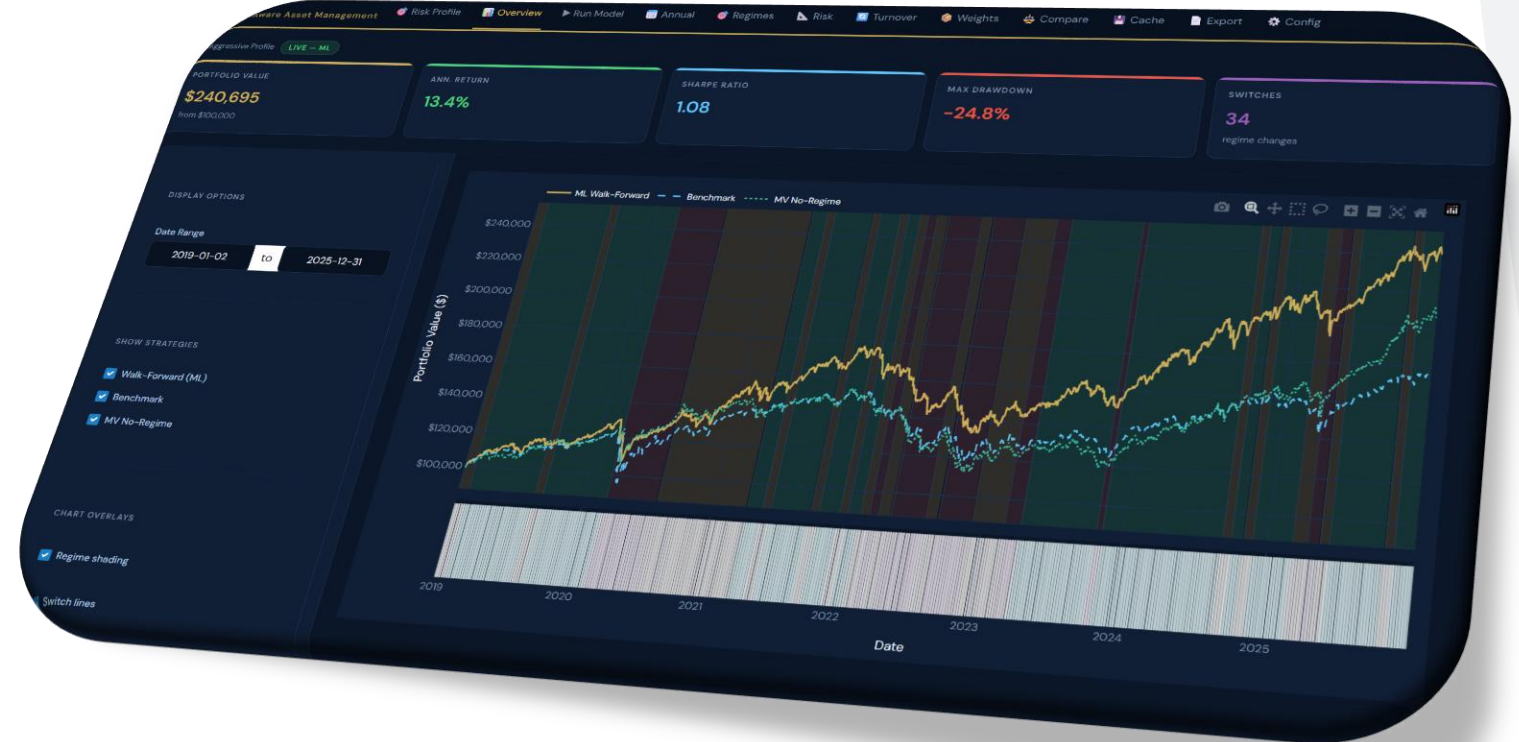
Models struggled as they
had no historical reference



Dashboard Demo – RAAM Platform

Shiny Dashboard – Python based

- Risk profiling survey
- Model selection and execution
- Performance results
- Regime attribution
- Current allocation recommendation



Risk Profiling – Survey Methodology

Input	Measure
Investment Horizon	How long the investor can stay invested
Reaction to a 25% Drop	Behavioral response to losses
Maximum Loss Tolerance	Hard floor on acceptable drawdown
Investment Style	Self-assessed risk preference
What Upsets You More	Loss aversion vs opportunity cost
Income Stability	Ability to absorb losses from income
Investment Knowledge	Confidence in financial decisions
Experience with Volatility	Comfort with market swings

Profile	Gamma Range	Benchmark
Conservative	High	30% Equity / 60% Bonds / 10% Gold
Moderate	Medium	55% Equity / 40% Bonds / 5% Gold
Aggressive	Low	85% Equity / 10% Bonds / 5% Gold



To Sum up

Regime detection works ; the question is not whether to use it, but which model fits your investor's profile best.

- **Key Takeaways:**

- Regime-aware methodology is effective, both ML and Jump models generated consistent alpha across different market conditions and different risk profiles, outperforming the static benchmark in 6 out of 7 years
- ML and Jump were the best performing models overall/ ML delivered the most normally distributed returns, best drawdown control, and most consistent alpha across all three regimes. Jump had the lowest turnover with only 8 switches over 7 years
- Model choice depends on the investor; ML is better for capital preservation and downside protection. Jump is better for cost efficiency and long-term stable allocation

- **Looking Forward:**

- As more data becomes available, the models will learn new regimes such as the 2022 simultaneous equity and bond decline and allocate accordingly





Q&A

Thank you